

Predicting the Length of Stay of Animals from Animal Shelter Data

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Introduction

Motivation:

There are millions of animals entering shelters and rescues in the U.S. While shelter staff often rely on their intuition and experience for decision-making.

Problem:

What kind of animals are the most vulnerable and how can we help improve their outcome?

Objective:

Using a data-driven approach to provide a consistent, evidence-based insights for animal outcomes, helping shelters achieve their ultimate goal of saving lives.

Solution:

Using machine learning models to predict their length of stay in the shelter.

Data Overview

- **Source:** Austin Animal Center open data portal (May 2013 – May 2025)
- Two tables: Intakes (173,812 rows) & Outcomes (173,335 rows)
- **Challenge:** Animals can appear multiple times across years
- **Solution:** Treated each intake-outcome pair as an "episode" (visit)
- **Episode construction process:**
 - Paired chronologically adjacent intake/outcome events per animal
 - Removed exact duplicates
 - Corrected timestamp mis-orderings
- **Final dataset:** 172,800 visits

Feature Engineering

Target Variable:

Length of Stay (LOS): days
between intake and outcome

Categories	Features
Basic Info	Age (in days), gender, has name, mixed vs purebred
Calendar	Intake month, day of week, season
Condition	Medical condition, spay/neuter status
High-Cardinality	Breed → "Top-K + Other" grouping Color → Base colors + pattern flags

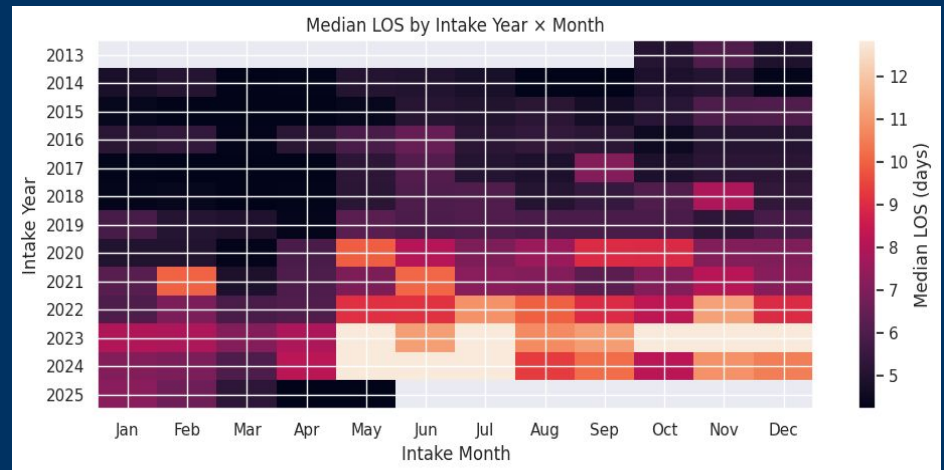
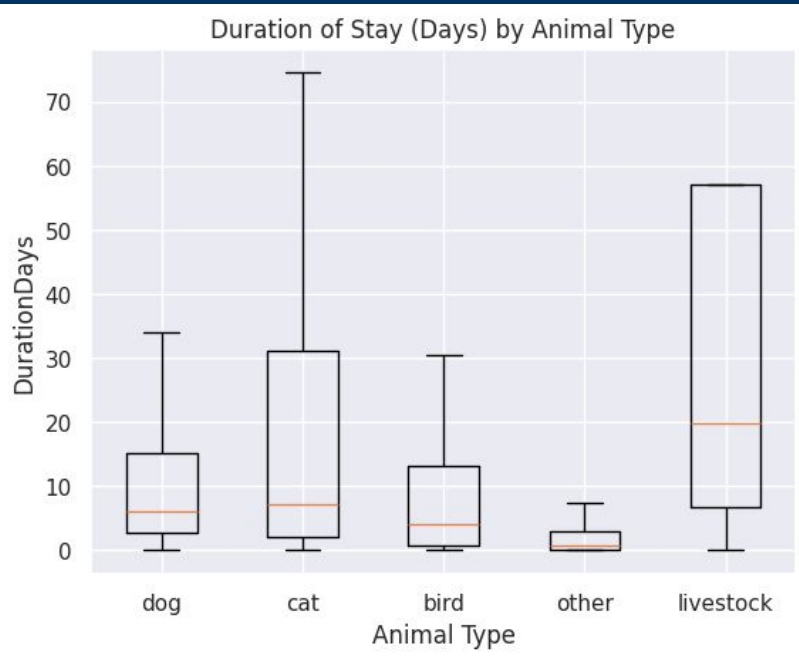
Key EDA Insights

Key Findings:

- Dogs + cats = 94.3% of visits → focused modeling on these species
- LOS is heavily right-skewed:
 - Median: 6 days
 - Mean: 21 days
 - ~18% are long-stay outliers (important for capacity planning)

Strong temporal patterns:

- LOS drifting upward over time (step change ~2021)
 - Strongest signal by year, not month
- Motivated calendar features & time-based data splits



Train / Validation / Test Split

- Why temporal? Simulates real-world deployment
 - Model must predict FUTURE outcomes from PAST patterns
- Final feature counts:
 - Cats: 80 features
 - Dogs: 146 features (more breed diversity)

2013-2019	2020-2022	2023-2025
Training	Validation	Test
66%	19%	15%

Methods

Linear regression:

- strong interpretability
- computationally efficient
- optimization: remove outliers, learning rate, batch size, number of epochs

Tree-based model:

- handle categorical features effectively
- mitigate overfitting
- optimization: number of estimators, maximum depth, and learning rates.

Neural network:

- learns complex non-linear relationships
- well-suited for high-dimensional feature spaces
- optimization: architecture depth, dropout rate, L2 regularization, learning rate, batch size

Evaluation:

Root Mean Square Error (RMSE)

Mean Absolute Error (MAE)

Model | Linear Regression

Hyperparameter Tuning:

- Learning rate and number of epochs

Result:

All Data(days)				Remove outliers(days)		% Improvement	
Metrics	Mean	RMSE	MAE	RMSE	MAE	RMSE	MAE
Dog	68.10	60.86	31.16	12.48	9.01	79.5%	71.1%
Cat	39.41	41.87	24.78	27.28	19.11	34.8%	22.9%

Model | Tree-based Model

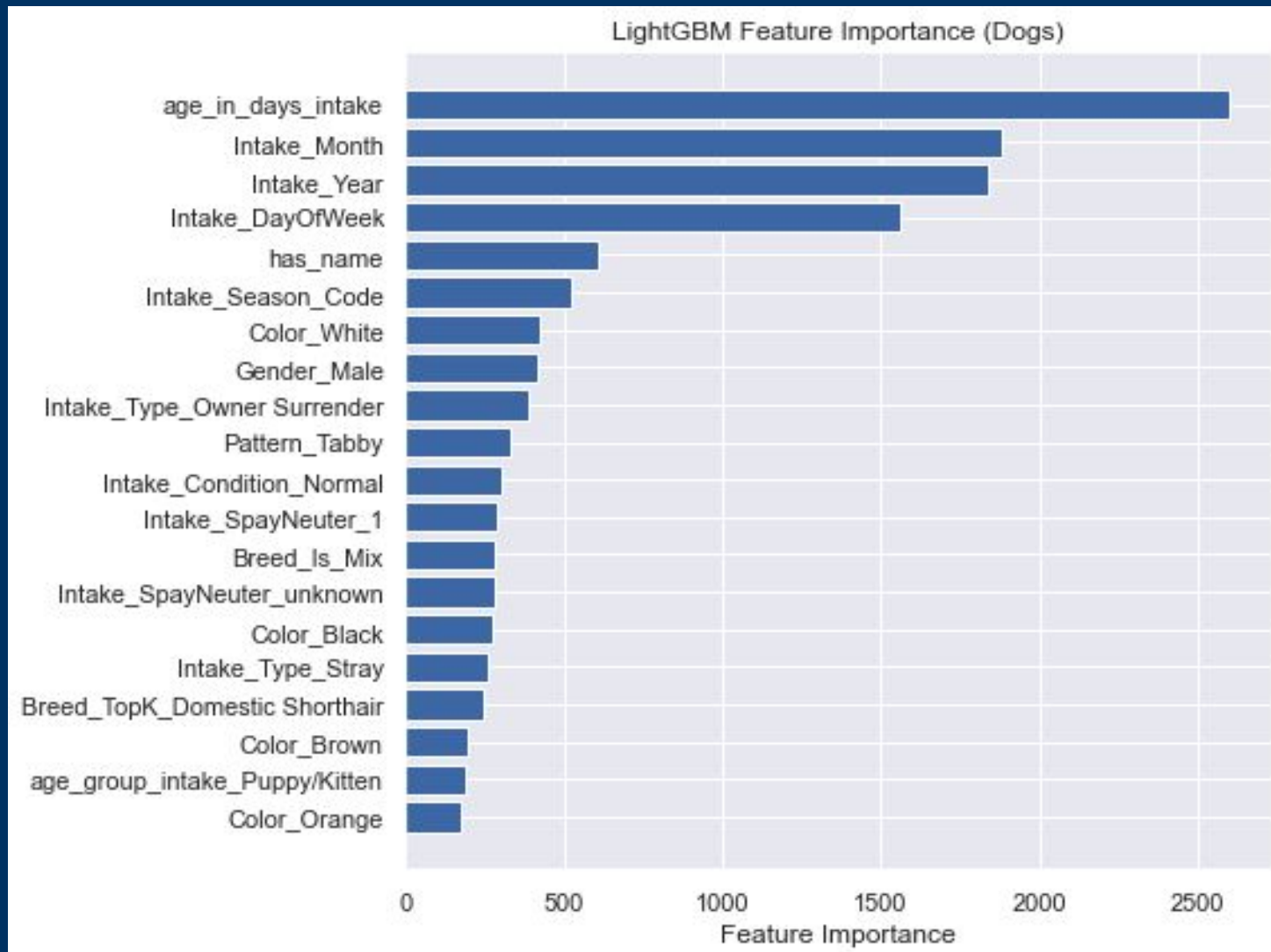
Hyperparameter Tuning:

- number of leaves, learning rate, maximum depth, subsample ratio, columns sampling ratio, minimum child samples

Result:

Metrics	Baseline		LightGBM(best)		% Improvement	
	RMSE	MAE	RMSE	MAE	RMSE	MAE
Dog	12.48	9.01	8.62	6.27	30.9%	30.52%
Cat	27.28	19.11	17.37	12.96	36.32%	32.20%

Model | Tree-based Model



Model | Neural Network

Hyperparameter Tuning:

- Tested 15 configurations per species
- Varied: architecture depth, dropout, L2, learning rate, batch size

Optimal Architectures (species-specific):

- Cats: simpler [64, 32] network
- Dogs: wider [256, 128, 64] network

Results:

	Baseline		Improved		% Improvement	
Metrics	RMSE	MAE	RMSE	MAE	RMSE	MAE
Dog	51.78	25.39	6.77	5.09	87%	80%
Cat	33.4	21.81	16.60	12.46	50%	43%

Conclusion

Best performance model:

NEURAL NETWORK

Key Findings:

- Non-linear models significantly outperformed linear regression
- Dogs easier to predict than cats (more consistent patterns)
- IQR outlier removal critical for all models

Limitations:

- Excludes long-stay animals (8-14%) — the ones needing most intervention
- Limited to intake-time features only

Future Work:

- Ensemble methods (LightGBM + Neural Net)
- Computer vision for animal photos
- Two-stage model: classify short/long stay → specialized predictions

**Last,
If you ever consider to adopt an animal...**

Here is the website of Austin Animal Center:

<https://www.austintexas.gov/services/adopt-a-pet>

And...

